The file CA.txt contains census information on the frequency of given names in the period 1910-2017 for the US state of California. There are in excess of 370000 records in the file. Each record in the file has the following format: a time zone (either UTC-7 or UTC-8) gender (M = male or F = female), 4-digit year of birth (starting with 1910), the 2-15 character name, and the number of occurrences of the name. The file is sorted first on gender, then year of birth, and then on number of occurrences in descending order. When there is a tie on the number of occurrences, names are listed in alphabetical order. To safeguard privacy, there is a restriction on those names with less than five occurrences in a given year. The first record in the file is as follows:

UTC-8 F 1910 Mary 295

You must execute the following two tasks concurrently (with all tasks submitted to the executor service). Importantly, you must read from the file once.

Task ONE

Write and execute a Callable that will accept three arguments.

1. A given name (either male or female). Call this parameter *N.*
2. A year. Call this parameter *Y1*.
3. Another year. Call this parameter *Y2*.

The Callable will then determine and return the percentage difference between the number of people called *N* the year *Y1* versus the number of people called *N* in the year *Y2*.

To calculate the percentage difference: First: work out the difference (increase) between the two numbers you are comparing. Then: divide the increase by the original number and multiply the answer by 100. If your answer is a negative number then this is a percentage decrease.

Task TWO

Write and execute a Runnable that will accept a given name as an argument (either male or female) and will then print the number of times the name is found (using a **whole word search**) in the file along with its frequency for each year that it is listed. For example, the following would be the output for "Zack".

"Zack" was found 40 times.

1963 5

1969 5

1972 5

Etc……

Marks	
Read from the file **once**	15 Marks
Task 1	45 Marks
Task 2	40 Marks